

BTR REFORGED DRIVER ADS VALIDATION

Proactive validation of the current BTR_CLI
feedback-stream artifact on Microsoft
Defender remediation drivers.

VALIDATION QUERY

CONFIDENCE: MEDIUM

SOURCE-CONFIRMED BEHAVIOR

retain facts, isolate interpretation

WindowsEvent

direct telemetry boundary

HUNT · VALIDATE · TUNE

no automatic malicious verdict

REPORT DATE

24 August 2026

VERSION

v1.0

PRIMARY TELEMETRY

Microsoft Sentinel ·
WindowsEvent

ATT&CK

T1564.004

Evidence-led dossier. Query status remains an untested
implementation sketch until executed in the target environment.
INDEPENDENT RESEARCH FORMAT · CONTROLLED UI/UX MASTER



EXECUTIVE ASSESSMENT

Decision-grade summary for detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Validate the current PoC differentiator; do not alert on legitimate BTR behavior.

Check Point reproduced a BTR_CLI-specific secondary .dat Alternate Data Stream on the staged .sys driver. The universal :changelist stream, random service and Microsoft signature are not discriminators because legitimate Defender remediation produces them too.

OPERATIONAL VALUE

Proactive coverage before observed real-world weaponization.

The query provides a narrow experiment for a signed kernel primitive that can delete protected security files and Registry keys. It is deliberately a Validation query because no in-the-wild abuse was observed and the current PoC artifact is mutable.

9.2

01

OPERATIONAL VALUE



Medium

02

DETECTION CONFIDENCE



Validation

03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACTS

Observed behavior

- Check Point reverse engineered BTR.sys and reproduced file and Registry operations from kernel mode.
- Sysmon Event ID 15 captured BTR_CLI feedback in a secondary .dat ADS on the .sys driver.
- Legitimate MpEngine.dll wrote feedback as a standalone file under Windows Defender RebootActions; no real-world abuse was observed.

ANALYTICAL ASSESSMENT

What the evidence supports

- The secondary .dat ADS is the clearest current discriminator from the legitimate workflow.
- A .sys:changelist event alone is expected in both legitimate remediation and the PoC.
- A match warrants kernel-driver and privilege investigation but does not prove malicious operations.

DETECTION ASSUMPTIONS

What must be validated

- Sysmon Event ID 15 is enabled and collected from the Operational channel.
- AMA DCR or WEF places the event in WindowsEvent with EventData.TargetFilename populated.
- The current PoC retains its source-observed feedback-stream pattern.

Confidence boundary

This package detects neither BTR.sys generally nor successful Defender neutralization. It validates only the current PoC-specific .dat ADS on a .sys file. Legitimate Defender behavior, schema population and local enterprise overlap must be measured first.



THREAT MECHANICS

Source-observed chain and explicit observability boundaries.

ATTACK FLOW

01

Obtain privilege

The technique requires administrative access and SeLoadDriverPrivilege.

02

Stage BTR.sys

The tool extracts and registers the signed Defender remediation driver.

03

Write configuration

An encrypted :changelist ADS defines kernel file or Registry operations.

04

Write feedback

The current PoC places its report in a secondary .dat ADS on the driver.

05

Execute operation

The signed driver can delete, move or modify protected objects from kernel mode.

Observable core

A Sysmon FileCreateStreamHash event whose TargetFilename contains .sys: and ends in .dat, plus raw event, host and event identifiers.

Non-observable boundary

Malicious intent, transaction contents, successful kernel operations, Defender disablement and attacker identity from this event alone.

DETECTION HYPOTHESIS

A current BTR_CLI run should expose a secondary .dat ADS on its staged driver.

If an attacker uses the current BTR_CLI PoC to configure BTR.sys, Sysmon telemetry should show creation of a secondary .dat ADS on a .sys driver. The behavior can be distinguished from the legitimate Defender workflow observed by Check Point because legitimate feedback is written as a standalone file under RebootActions.

REQUIRED TELEMETRY

WindowsEvent

- Sysmon Event ID 15, FileCreateStreamHash.
- Microsoft-Windows-Sysmon/Operational collected into WindowsEvent.
- EventData.TargetFilename plus stable host and time fields.

OPTIONAL ENRICHMENT

Context that improves disposition

- Sysmon Event IDs 6, 12, 13 and 23.
- Windows System Event ID 7045 and Defender process context.
- Privilege, service Registry and reboot timeline evidence.

UNAVAILABLE HERE

Do not infer

- Transaction intent or RC4 payload interpretation from the filename predicate.
- Proof of kernel-operation success.
- Reliable attribution to the initiating user-mode tool from the ADS event alone.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and quality gates for a defensible hunt.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Sentinel · WindowsEvent

Microsoft documents Sysmon Event ID 15 for named stream creation and WindowsEvent for custom Windows Event Log collection. Check Point provides the exact source-observed TargetFilename distinction.

EVIDENCE BOUNDARY

A matching event is a lead, not a verdict.

EventData is documented as a dynamic property bag; TargetFilename is present in Sysmon Event ID 15 but must be validated in the target connector output. The query keeps RawEventData for schema troubleshooting.

REQUIRED AND ENRICHMENT FIELD MODEL

CATEGORY	FIELD	PURPOSE	REQUIREMENT
Log event	WindowsEvent	Custom Windows Event Log records.	REQUIRED
Source	Provider, Channel, EventID	Sysmon Operational Event ID 15.	REQUIRED
Artifact	EventData.TargetFilename	Named stream target from Sysmon.	REQUIRED
Time	TimeGenerated	Event-time investigation pivot.	REQUIRED
Host	Computer, _ResourceId	Asset and collection context.	REQUIRED
Raw evidence	EventData, RawEventData	Parsing and field-validation evidence.	OPTIONAL
Correlation	EventRecordId, Correlation	Event-level investigation context.	OPTIONAL

Data quality requirements

- Confirm the Sysmon Operational channel is included in the AMA DCR or WEF subscription.
- Create an approved benign .sys:.dat ADS in an isolated lab and retain the raw Event ID 15.
- Verify EventData.TargetFilename population and exact Provider and Channel values.
- Run a controlled Defender reboot-remediation workflow to measure legitimate overlap.

Known failure conditions

- Sysmon Event ID 15 is disabled or filtered.
- The Sysmon Operational channel is not forwarded to WindowsEvent.
- EventData parsing omits TargetFilename or only RawEventData is retained.
- The PoC changes its feedback location or the artifact predates retention.



DETECTION LOGIC

Primary-platform logic with every condition traceable to evidence.

VALIDATION QUERY

DETECTION OBJECTIVE

Verify that the current BTR_CLI feedback-stream signal is collectable and distinguishable.

Establish field population, controlled positive coverage and benign Defender overlap before considering an alert.

ANALYTICAL LOGIC

Direct filters and investigable output.

Direct filter on Sysmon provider, Operational channel, Event ID 15 and the source-observed .sys:.dat stream shape. No threshold, rarity, join, sequence, signer assumption or malicious verdict.

KQL · UNTESTED VALIDATION

```
// Validation query - Untested implementation sketch
// Primary platform: Microsoft Sentinel / Log Analytics
// Prerequisite: Sysmon Event ID 15 from Microsoft-Windows-Sysmon/Operational
// collected into WindowsEvent through an AMA DCR or Windows Event Forwarding.
// Evidence: Check Point observed BTR_CLI writing its feedback report to a
// secondary .dat ADS on the .sys driver. The current legitimate Defender
// workflow writes feedback as a standalone file under RebootActions instead.
WindowsEvent
| where Provider == "Microsoft-Windows-Sysmon"
| where Channel == "Microsoft-Windows-Sysmon/Operational"
| where EventID == 15
| extend TargetFilename = tostring(EventData.TargetFilename)
| where TargetFilename contains ".sys:"
  and TargetFilename endswith ".dat"
| project
  TimeGenerated, Computer, EventRecordId, Provider, Channel, EventID,
  TargetFilename, EventData, RawEventData, SystemProcessId, SystemUserId,
  Correlation, _ResourceId
| order by TimeGenerated desc
```

STATUS	THRESHOLD	JOIN / SEQUENCE	PRIMARY KEY	DISPOSITION
Untested validation	None	None	Computer + EventRecordId + TargetFilename + TimeGenerated	Manual triage

What a match means

A .dat named stream was created on a .sys file and requires raw-event, driver and privilege investigation.

What a match does not prove

BTR.sys abuse, successful kernel actions, Defender neutralization or attacker presence.



TRIAGE & INVESTIGATION

Evidence collection before escalation.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Preserve the raw event.
Capture EventData, RawEventData, host, time and record identifiers.
- 02

Resolve the driver.
Hash and verify the base .sys file, service key, Args value and signature.
- 03

Establish lineage.
Review Sysmon driver-load, Registry and file-delete events plus System 7045.
- 04

Check privilege context.
Identify accounts and processes with SeLoadDriverPrivilege around staging.
- 05

Assess impact.
Inspect security-service keys, binaries, BootClean.log and reboot activity.

Escalation gate

Escalate when the .sys:.dat ADS aligns with non-Defender staging, direct Registry service creation, NtLoadDriver-like activity, security-file deletion, Defender service-key removal or unexplained privileged access.

Primary benign overlap

Authorized security research or another product may create a .dat ADS on a .sys file. Legitimate Defender currently writes feedback as a standalone file according to the source, but that behavior must be revalidated after platform updates.

INVESTIGATION PIVOTS

TargetFilename
Base driver path and secondary stream name.

Computer
Role, Defender health and Sysmon coverage.

Driver load
Sysmon Event ID 6 and signature/hash evidence.

Registry
Service key, Group, Args and Start values.

File deletes
System PID 4 activity against security binaries.

Privilege
Assignment and use of SeLoadDriverPrivilege.

DISPOSITION CRITERIA

OUTCOME	MINIMUM EVIDENCE	ACTION
BENIGN	Approved, owned and repeatable activity.	Document and tune narrowly.
SUSPICIOUS	Unknown activity without corroborated malicious follow-on.	Deepen endpoint review.
LIKELY MALICIOUS	Unauthorized match plus coherent file, process, network or control-plane evidence.	Escalate incident response.



FALSE POSITIVES & VALIDATION

Operational controls required before promotion.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	WHY IT OVERLAPS	TUNING ACTION
Security validation	A lab may intentionally reproduce the documented ADS pattern.	Require approved host, operator and test window.
Forensic tooling	Responders may preserve or recreate named streams.	Validate case ownership and acquisition provenance.
Other endpoint software	A product may use ADS metadata on driver files.	Confirm signer, owner, product documentation and stable path context.

Baseline plan

- Run across available WindowsEvent retention after enabling Event ID 15.
- Review every .sys:.dat match with endpoint and Defender owners.
- Measure TargetFilename parsing and enterprise ADS-producing software.
- Do not add a count, rarity or time-window threshold.

Validation procedure

- Generate the controlled positive and two negative controls.
- Compare local Sysmon XML with WindowsEvent.EventData and RawEventData.
- Exercise an approved Defender reboot-remediation case and confirm no false match.
- Review any historical matches with driver, Registry and privilege evidence.

TEST CASES

CASE	EXPECTED RESULT
Create an approved .sys:report.dat ADS in an isolated lab	Expected match if Event ID 15 collection and parsing work.
Create .sys:changelist without a .dat stream	No match; this is universal BTR behavior.
Create a standalone .dat file under RebootActions	No match by design.
Observe local Event ID 15 but no WindowsEvent record	Failure: repair Sysmon collection before rule design.

Hunting

Initial query

Candidate

Current status

Production

Measured precision

Optimized

Stable feedback

Success

The controlled .sys:.dat event matches, negative controls do not, TargetFilename is stable, and legitimate Defender remediation does not produce the same signal.

Rejection

Legitimate enterprise software frequently creates the same pattern and no durable driver, lineage or ownership discriminator can be established.

Failure

The controlled event is absent, truncated or lacks a usable TargetFilename field.



ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

MITRE ATT&CK MAPPING

TECHNIQUE	NAME	COVERAGE STATEMENT
T1564.004	Hide Artifacts: NTFS File Attributes	Directly covers the use of an NTFS Alternate Data Stream for PoC feedback data. It does not claim coverage of the kernel file and Registry operations.

Detection coverage summary

Strong for the current PoC feedback pattern when Sysmon Event ID 15 is collected; absent for changed feedback paths, cleared logs, disabled Sysmon and historical activity outside retention.

Residual risk

An operator can modify the stream name or feedback destination, use the universal legitimate-looking :changelist path only, disable collection or abuse another signed primitive. Privilege controls and broader driver lineage remain mandatory.

ANALYST CONCLUSION

A high-value proactive validation with an explicit non-malicious verdict.

Primary research and official telemetry support a narrow, testable signal. Environment-specific EventData parsing, possible benign ADS usage and the lack of observed attacker deployment prevent High confidence or Production classification.

REFERENCES

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DOCUMENT CONTROL

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Publication control

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